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Lung cancer screening use among screening-eligible adults with disabilities

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Abstract

Background: Lung cancer screening (LCS) use among adults with disabilities has not been well characterized. We estimated the prevalence of LCS use by disability types and counts and investigated the association between disability counts and LCS utilization among LCS-eligible adults.

Methods: We used cross-sectional data from the 2019 Behavioral Risk Factor Surveillance System, Lung Cancer Screening Module. Based on the 2013 US Preventive Services Task Force criteria for LCS, the sample included 4407 LCS-eligible adults, aged 55–79 years, with current or former (quit smoking in the past 15 years) tobacco use history of at least 30 pack-years. Disability types included limitations in hearing, vision, cognition, mobility, self-care, and independent living. We also categorized respondents by number of disabilities (no disability, 1 disability, 2 disabilities, 3+ disabilities). We utilized descriptive statistics and multivariable logistic regression analyses to determine the association between disability counts and the receipt of LCS (yes/no) in the past 12 months.

Results: In 2019, 16.4% of LCS-eligible adults were screened for lung cancer. Overall, 49.6% of participants had no disability, and 14.5% had >3 disabilities. Mobility was the most prevalent disability type (35.4%), followed by cognitive impairment (18.2%) and hearing (16.6%). LCS was

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AUTHOR CONTRIBUTIONS

HP, CP, IR: Study concept and design. HP: Acquisition of subjects and/or data. HP, IR. SS, CP: Analysis and interpretation of data. HP, IR, SS, CP: Preparation of manuscript.

CONFLICT OF INTEREST STATEMENT

None to report.

SPONSOR'S ROLE

The sponsor had no role in the study.

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more prevalent in adults with disability in self-care versus no disability in self-care (24.0% vs. 15.5%, $p = 0.01$), disability in independent living versus no disability in independent living (22.2% vs. 15.4%, $p = 0.02$), and cognitive impairment disability versus no cognitive impairment (22.1% vs. 15.3%, $p = 0.03$). The prevalence rates of LCS among groups of LCS-eligible adults with different disability counts were not significant ($p = 0.17$).

Conclusions: Despite the lack of clinical guidelines on LCS among individuals with disabilities, some individuals with disabilities are being screened for lung cancer. Future research should address this knowledge gap to determine clinical benefit versus harm of LCS among those with disabilities.

Keywords

disability counts; disability types; LDCT screening; lung cancer

INTRODUCTION

Lung cancer is the leading cause of cancer-related deaths in the United States.[1] In 2023, an estimated 238,340 new patients were diagnosed, and 127,070 Americans died from lung cancer.[1] Although the treatment options for lung cancer have improved, the disease still has a poor survival rate, with only up to 25% of diagnosed patients surviving 5 years.[2,3] However, when lung cancer is diagnosed at an early stage, many treatment modalities are available that can improve overall survival.[4–7] Unfortunately, only 26% of patients are diagnosed at an early stage when the 5-year survival rate is 61%.[3] Thus, screening, and early diagnosis efforts are crucial steps for improving the long-term prognosis for those at increased risk of developing lung cancer.

Annual lung cancer screening (LCS) with low-dose computed tomography (LDCT) decreases lung cancer mortality up to 39%.[8–10] The US Preventive Services Task Force (USPSTF) currently recommends annual LCS for high-risk individuals—those between 50 and 80 years of age who are current and former (quit within past 15 years) smokers with at least a 20 pack-year smoking history.[11] Since 2015, the Centers for Medicare & Medicaid Services (CMS) has provided coverage for annual LCS with LDCT to appropriate beneficiaries who meet eligibility requirements.[12] Despite the LCS recommendation and related coverage decisions, screening with LDCT is underutilized. In fact, less than 16% of eligible adults receive LCS with LDCT.[13]

Research shows that individuals with disabilities are even more vulnerable to lung cancer as they have worse outcomes than those without disabilities.[14] For example, findings from a large-scale study reported a higher mortality rate among patients with lung cancer and disabilities compared to their counterparts without disabilities.[14] Yet limited evidence exists regarding whether adults with disabilities who are eligible for LCS actually receive LCS. A study that investigated the use of cancer screening services for people with functional disabilities (hearing, vision, and mobility) and cognitive impairment reported significant differences in utilization of such services. Specifically, people with functional disabilities were less likely to be screened for breast and cervical cancers compared to people without functional disabilities.[15] Similarly, a recent study showed that women with a physical

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disability were 17% less likely to be screened for cervical cancer compared to women without a physical disability.[16] Limited evidence exists about LCS use among screening-eligible individuals with disabilities. Thus, understanding LCS use among people with disabilities is critically needed. In this study, we (1) estimated the prevalence of disability types and counts, (2) estimated the prevalence of LCS use by disability types and counts, and (3) investigated the association between disability counts and LCS utilization among screening-eligible adults living in 16 US states in 2019.

METHODS

Data source

Cross-sectional data from the 2019 Behavioral Risk Factor Surveillance System (BRFSS), Lung Cancer Screening Module (LCSM) were used. BRFSS is a population-based, nationally representative telephone survey of noninstitutionalized adults aged 18 years and older. Established in 1984 with 15 states, BRFSS is now conducted annually by state health departments in collaboration with the Centers for Disease Control and Prevention (CDC) in all 50 US states and the District of Columbia, and three US territories, including Puerto Rico, Guam, and the US Virgin Islands; it is the largest continuously conducted health survey system in the world.[17] The BRFSS questionnaire consists of three parts: the core questionnaire, optional modules on specific topics, and state-added questions. All US states, District of Columbia, and US territories must administer the core questionnaire without modification each year. However, states may choose to administer optional modules and state-added questions according to their needs. The core questionnaire collects information on population socio-demographics, chronic health conditions, health status, health-related risk behaviors, healthcare access, and preventive service use. The total sample size for the 2019 BRFSS was 418,268, and the median response rate was 49.4%.[18]

The LCSM is an optional module that was administered by 16 US states in 2019 (Arizona, Idaho, Kentucky, Maine, Minnesota, Missouri, Montana, North Carolina, North Dakota, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, West Virginia, and Wisconsin). It collects data on age when participants first started to regularly smoke cigarettes, age when participants last smoked cigarettes regularly, average number of cigarettes smoked per day, and whether the participant had a CT scan in the past year to check for lung cancer. The response rates for the 16 states that administered the LCSM in 2019 ranged from 40.8% in North Carolina to 60.8% in North Dakota.[18] Because we used publicly available and de-identified data, no institutional review board approval was needed in accordance with the exempt criteria under 45 Code of Federal Regulations part 46.102.[19]

Study population

We included screening-eligible adults living in those 16 states that administered the LCSM. Eligibility criteria were based on USPSTF initial criteria as we used the 2019 data. The current criteria, which were established in 2021, did not apply. LCS criteria were updated in March 2021 to decrease the starting age for screening from 55 to 50 and smoking history from 30 pack-years to 20 pack-years.[11,20,21] Screening-eligible participants considered for this study were those aged 55–79 years old who were current and former smokers (had

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quit within the past 15 years) with at least 30 pack-years of smoking history. We calculated smoking pack-year history based on the National Cancer Institute's definition using the following formula: (age at which participants reported last smoking cigarettes regularly—age at which participants started smoking cigarettes regularly) × (number of cigarettes participants smoked each day on average/20).[22] Those who had a smoking pack-year history of less than 30 were excluded from this study. Those who quit smoking for more than 15 years were also excluded. After removing observations with missing values, the analytic sample (Supplementary Figure S1) included 4407 LCS-eligible participants as per the USPSTF guidelines.

Study variables

Outcome variable—The primary outcome variable was self-reported LCS in the past 12 months (yes/no). Participants were asked the following: "In the last 12 months, did you have a CT or CAT scan?" The response options were: 1 = "yes, to check for lung cancer," 2 = "no (did not have a CT scan)," and 3 = "had a CT scan, but for some other reason." Participants who reported that they had a CT scan, but for some other reason were recoded as not having a CT scan for LCS. Those who reported "don't know/not sure" or had a missing answer were excluded.

Key independent variable—The key independent variable was self-reported disability status. BRFSS uses the 6-item set of questions from the American Community Survey to measure disability status. These items are deemed the minimum data standard mandated by federal statute.[23–25] The six items assessed the following disability types: hearing, vision, cognition, mobility, self-care, and independent living. Participants were asked the following six questions, and response choices were "yes" or "no."

1. Are you deaf, or do you have serious difficulty hearing? (hearing)
2. Are you blind, or do you have serious difficulty seeing, even when wearing glasses? (vision)
3. Because of a physical, mental, or emotional condition, do you have serious difficulty concentrating, remembering, or making decisions? (cognition)
4. Do you have serious difficulty walking or climbing stairs? (mobility)
5. Do you have difficulty dressing or bathing? (self-care)
6. Because of a physical, mental, or emotional condition, do you have difficulty doing errands alone such as visiting a doctor's office or shopping? (independent living)

Participants were considered to have the particular disability types if they responded "yes" to the corresponding question.[23–25] For this study, we combined disabilities and categorized disability status into four groups: 1) no disability; 2) 1 disability; 3) 2 disabilities; and 4) 3 or more disabilities. We also investigated disability types separately as six binary variables with response options of "yes" or "no."

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Covariates—We included the following self-reported demographic characteristics of participants: age, sex, race and ethnicity, and marital status. Socioeconomic characteristics included educational attainment, household annual income, health insurance coverage, US census region, metropolitan residency, and unmet medical care needs due to cost in the past 12 months. We included the following clinical characteristics: self-reported general health status; history of cancer; history of asthma; history of chronic obstructive pulmonary disease, emphysema, or chronic bronchitis; and time since last routine health checkup.

Statistical analyses

We applied the CDC's recommended sampling weights to account for the BRFSS's complex survey design to generate population estimates. BRFSS developed two steps for the weighting procedure: design weighting, which reduces bias because of unequal probability of participant selection, and iterative proportional fitting or raking, which adjusts for demographic differences between those individuals who are sampled and the population that they represent. BRFSS used age, sex, race, marital status, Hispanic ethnicity, education, home and phone ownership, and state region variables in the raking process.[26,27]

We conducted descriptive statistics, including frequencies, means, and percentages, with corresponding 95% confidence intervals (CI) and standard errors (SE) of study variables to describe the sample overall and by disability status. We conducted a 3-step multivariable logistic regression analyses to investigate the association between disability status and LCS use in the past year by sequentially controlling for participants' demographic, socioeconomic, and clinical characteristics using the following approach: model 1 included only demographic characteristics; model 2 included variables from model 1 and socioeconomic characteristics; model 3 included variables from model 2, clinical variables, and smoking status. We report an estimated adjusted odds ratios (OR) and 95% CI. We used the *svy* command in Stata 17 (StataCorp LLC, College Station, TX) to conduct weighted data analysis. We considered a two-tailed p-value of <0.05 to be statistically significant.

RESULTS

Demographic and clinical characteristics of study participants

A total of 4407 LCS-eligible participants were included in the sample, corresponding to a population estimate of 1,542,873 LCS-eligible individuals living in 16 US states. As presented in Table 1, most of those in the sample were male (57.6%) and self-reported as non-Hispanic White adults (90.0%). Over half (56.1%) had less than high school education and 34.2% were in the lowest income tier (<$25,000). Among participants, 6.8% did not have health insurance coverage and 13.0% reported unmet healthcare needs due to cost in the past 12 months. Overall, 51.7% of participants were current smokers and 48.3% were former smokers with a mean pack-year smoking history of 53.2 years (SE 0.71). In 2019, a total of 16.4% of LCS-eligible individuals reported undergoing LCS in the past 12 months (Table 1).

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Lung cancer screening by disability types and count

Overall, 49.6% of participants reported no disability, 23.6% of participants reported 1 disability, 12.3% reported 2 disabilities, and 14.5% had 3 or more disabilities. Mobility was the most prevalent disability type (35.4%), followed by cognitive disability (18.2%), hearing (16.6%), lack of independent living (13.7%), self-care (9.8%), and vision (7.8%) (Table 2). In 2019, 21.2% of LCS-eligible individuals with cognitive disability had LCS in the past 12 months compared to 15.3% without cognitive disability ($p = 0.03$) (Table 2 and Figure 1). LCS was more prevalent among those who had disability with self-care (24.0%) compared to screening-eligible individuals without self-care disability (15.5%, $p = 0.01$). LCS prevalence was also higher among participants who had disability in independent living (22.2%) compared to those who did not have disability in independent living (15.4%, $p = 0.02$). In 2019, 21.2% of individuals who reported three or more disabilities had LCS in the past 12 months (Table 2 and Figure 2). The prevalence rates of LCS among groups of LCS-eligible adults with different disability counts were not significant ($p = 0.17$) (Table 2).

Association between disability count and LCS

The results of multivariable regression models are presented in Table 3 and Supplementary Figure S2. There was a statistically significant association between disability and LCS in models 1 and 2. Specifically, individuals with at least 3 disabilities had 69% increased odds of reporting LCS in the past 12 months compared to those without disability in both model 1 (OR = 1.69, 95% CI 1.14–2.49, $p = 0.008$) and model 2 (OR = 1.69, 95% CI 1.12–2.54, $p = 0.013$). Controlling for all covariates in the full model (model 3), there was no significant association between disability count and LCS use. In the full model, individuals with health insurance coverage, those who had a routine health checkup in the past year, those with lung disease, those with no unmet healthcare needs due to cost in last 12 months, and those with a cancer history had higher odds of receiving LCS compared to their counterparts; these relationships were statistically significant.

DISCUSSION

The results of this population-based study of adults eligible for LCS demonstrate that LCS is underutilized in the United States. Among LCS-eligible individuals, only 16.4% received LCS in the past 12 months. Our results also showed that individuals who had disabilities in self-care and independent living and those with a cognitive disability had a higher prevalence of LCS use compared to participants with no such disabilities. Our study findings are consistent with previous findings reporting higher prevalence rates of LCS among individuals with disabilities in self-care and independent living.[28] However, another study did not find a significant difference in LCS use between screening-eligible adults with and without functional disability.[29] To date, little work has been done to investigate the association between disabilities and LCS. Prior research demonstrates that women with mobility limitations had a lower prevalence rate of receiving a Papanicolaou test for cervical cancer screening than women without mobility limitations. Additionally, women with cognitive impairment had a lower prevalence of mammograms compared to women without cognitive impairment.[15] This work provides evidence about the importance

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of studying disability types and cancer screening use. Therefore, more studies are needed to explore the relationships between disability types and receipt of LCS.

Our study findings also showed that participants with multiple disabilities (3+ disabilities) were more likely to be screened for lung cancer compared to those without disability. However, in the full model after we adjusted for clinical variables such as a history of asthma or other lung disease, the association between disability status and LCS use was no longer statistically significant. These findings suggest that the presence of concurrent lung disease may mediate the apparent relationship between LCS and disability. One explanation might be that patients with lung disease may be more likely to see a pulmonologist who, in turn, may be more likely to refer patients for LCS. Indeed, some studies have indicated that pulmonologists are more likely to be familiar with LCS and have fewer perceived barriers to ordering LCS than general practitioners.[30] Thus, contact with a pulmonologist may facilitate lung screening. However, current evidence on LCS use among individuals with lung disease and other comorbid conditions are mixed. One recent study conducted by Metwally and colleagues (2022) reported no statistically significant differences in LCS comparing individuals with and without self-reported lung-related comorbidity.[31] Several other studies have reported greater use of LCS among those with comorbid conditions.[28,29] Therefore, future studies are needed to enhance our understanding of the relationship between lung disease and LCS use.

An alternate explanation for our findings is that patients with poorer health and more disability may have more frequent interactions with healthcare providers that may facilitate conversations and referrals for LCS. Consistent with this hypothesis, our study showed that individuals who had routine checkups within the past year were more likely to undergo LCS, which is consistent with findings reported previously.[32]

Our study raises important questions about screening in populations that differ from those typically studied in clinical trials. In fact, participants in the National Lung Screening Trial were healthier and had a longer life expectancy than the general population of LCS-eligible adults.[33] However, this study highlights that lung diseases and disabilities are common among LCS-eligible patients. We estimated that 778,284 individuals eligible for LCS had at least 1 disability and 21.2% of adults with 3+ disabilities received LCS in the past 12 months. Current national guidelines developed by major organizations such as the National Comprehensive Cancer Network recommend against LCS for individuals with poor health conditions and impaired functional status.[34–36] However, impairments in functional status can vary widely and goals and preferences for medical care may also vary widely among people with disabilities. Future studies are needed to understand the benefits and risks of LCS among adults with disabilities that may aid with modifications of current clinical guidelines. Such evidence is critically needed to guide clinicians in their decision-making processes for LCS among individuals with disability.

Our study has several limitations that need to be acknowledged. The generalizability of study findings may be limited as not all states administered the LCSM. Selection bias might be another limitation that further limits the generalizability of the study findings. Specifically, people with disabilities that impair communication may be less

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likely to participate in the BRFSS and, therefore, their experiences with LCS may be underrepresented. The BRFSS survey collapses age for those 80 and older, and there is no way to separate participants who are 80 years old; therefore, we included individuals between 55 and 79 years old. The majority, 90%, of the sample was non-Hispanic White. Future studies with racially and ethnically diverse participants are critically needed. The items included in the LCSM have not been validated to our knowledge; future studies to validate them are needed. Disability status was measured by six items from the American Community Survey, which does not measure impairments comprehensively and might be a potential limitation of the current study. Therefore, future population-based studies are needed to assess disability status by employing well-developed, validated, and multi-item tools to measure disability. All variables in the current study are self-reported, so responses may be subject to recall bias. Despite these limitations, our study is strengthened by the use of a large, nationally representative population-based sample.

In conclusion, individuals with self-reported disability and cognitive impairment are being screened for lung cancer despite little evidence of the benefits of screening and lack of clinical guidelines on LCS among individuals with disabilities. These findings underscore the importance of conducting more research to help physicians, advanced practice providers, and patients with shared decision-making processes related to LCS. Evidence is also needed to help policymakers modify current practice guidelines. More research is also needed to understand the benefits of LCS among adults with disability, particularly among those with self-care, independent living, and cognitive disabilities.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Key points

- Annual lung cancer screening (LCS) with low-dose computed tomography decreases lung cancer mortality significantly, yet LCS is underutilized among screening-eligible adults in the United States.
- Disabilities are common among LCS-eligible adults.
- Those with self-care, independent living, and cognitive impairment disabilities had high rates of LCS use.

Why does this paper matter?

Individuals with disabilities are more vulnerable to lung cancer as they have worse outcomes than those without disabilities. However, limited evidence exists regarding whether adults with disabilities who are eligible for lung cancer screening (LCS) actually receive LCS. Our findings showed that despite a lack of clinical guidelines on LCS among individuals with disabilities, some individuals with disabilities are being screened for lung cancer. Our findings underscore the importance of conducting more research to help providers and patients with shared decision-making processes related to LCS that balance clinical benefit versus harm of LCS among those with disabilities.

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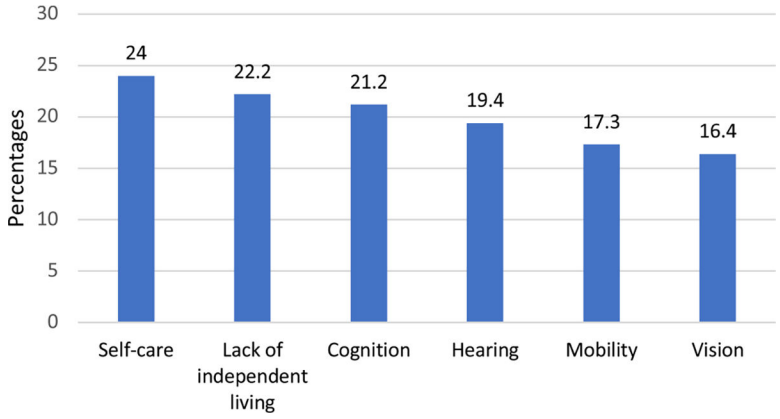


FIGURE 1.
Prevalence of lung cancer screening use by disability type.

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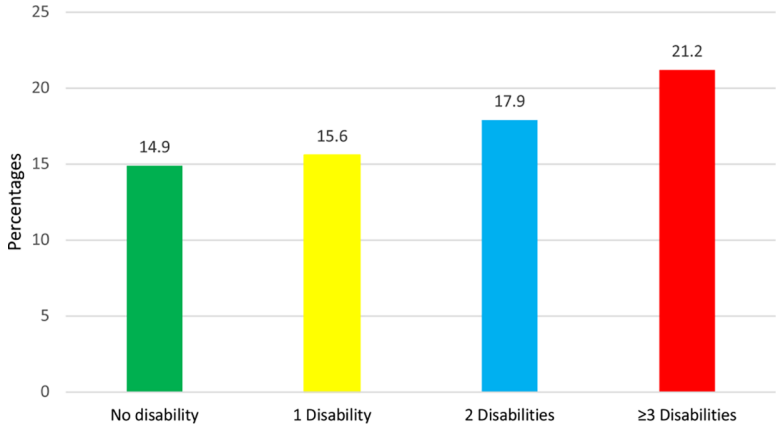


FIGURE 2.
Prevalence of lung cancer screening use by number of disabilities.

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TABLE 1

Demographic and clinical characteristics of adults at high risk for lung cancer.

Characteristics	Unweighted frequencie, *N* = 4407	Weighted % (95% CI), Weighted *n* = 1,542,873
Age		
55–64	2144	58.1 (55.3–60.9)
65–74	1855	35.6 (32.9–38.4)
75–79	408	6.3 (5.2–7.5)
Sex		
Male	2495	57.6 (54.7–60.3)
Female	1912	42.5 (39.7–45.3)
Race and ethnicity		
Non-Hispanic Black	114	5.7 (4.3–7.5)
Non-Hispanic White	4044	90.0 (88.0–91.7)
Other	205	4.3 (3.3–5.4)
Marital status		
Never	326	6.7 (5.5–8.2)
Married	2046	52.7 (49.9–55.5)
Divorced/Separated	1353	27.7 (25.2–30.3)
Widowed	682	12.9 (11.2–14.8)
Education		
Less than high school	2100	56.1 (53.4–58.8)
Attended college	1467	32.2 (29.6–34.8)
Graduated college	840	11.7 (10.4–13.1)
Income		
<$25,000	1675	34.2 (31.7–36.9)
$25,000 - < 50,000	1317	30.8 (28.1–33.5)
$50,000- < $75,000	688	16.0 (14.2–18.0)
$75+	727	19.0 (16.9–21.4)
Region		
Northwest	1283	26.0 (24.8–27.4)
Midwest	1461	30.9 (29.8–32.01)
South	919	31.8 (30.3–33.3)
West	744	11.3 (10.6–12.0)
Metropolitan residency		
Yes	2342	71.3 (69.1–73.4)
No	2065	28.7 (26.7–30.9)
Health insurance		
Yes	4146	93.2 (5.5–8.5)
No	261	6.8 (5.5–8.5)
Unmet healthcare need		
Yes	495	13.0 (11.3–15.0)
No	3912	87.0 (85.0–88.7)

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Characteristics	Unweighted frequencie, *N* = 4407	Weighted % (95% CI), Weighted *n* = 1,542,873
Routine check-up		
5+ years ago	196	5.2 (4.0–6.7)
Past year	3783	85.5 (83.4–87.4)
Past 2 years	308	7.1 (5.7–8.7)
Past 5 years	120	2.2 (1.54–3.1
Lung cancer screening		
Yes	777	16.4 (14.4–18.5)
No	3630	83.6 (81.5–85.6)
General health		
Fair-poor	1735	41.6 (38.8–44.4)
Very good-excellent	1189	24.5 (22.3–26.8)
Good	1483	33.9 (31.2–36.8)
Lung disease[a]		
Yes	1600	35.0 (32.4–37.7)
No	2807	65.0 (62.3–67.6)
Asthma		
Yes	670	14.4 (12.6–16.3)
No	3737	85.6 (83.7–87.4)
Cancer history		
Yes	712	16.4 (14.5–18.6)
No	3695	83.6 (81.5–85.5)
Smoking status		
Current	2143	51.7 (48.9–54.5)
Former	2264	48.3 (45.5–51.1)
	Weighted mean	**SE (95% CI)**
Age first started smoking (years)	16.30	0.12 (16.05–16.52)
Number of cigarettes per day	24.20	0.30 (23.56–24.76)
Years smoked	44.24	0.21 (43.82–44.65)
Pack-years smoking history	53.20	0.71 (51.78–54.55)

[a]Lung diseases include COPD, emphysema, and chronic bronchitis.

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TABLE 2

Lung cancer screening use by disability type and counts.

	Unweighted frequencies	Weighted % (95% CI)	LCS use	No LCS use	
Disability type	***N* = 4407**	***n* = 1,542,873**	**Weighted % (95% CI)**	**Weighted % (95% CI)**	***p*-value**
Hearing					0.20
Yes	783	16.6 (14.7–18.7)	19.4 (14.8–24.9)	80.6 (75.1–85.2)	
No	3624	83.4 (81.3–85.3)	15.8 (13.7–18.1)	84.2 (81.9–86.4)	
Vision					0.98
Yes	385	7.8 (6.5–9.4)	16.4 (11.3–23.2)	83.6 (76.8–88.7)	
No	4022	92.1(90.6–93.5)	16.4 (14.3–18.7)	83.6 (81.4–85.7)	
Cognitive					
Yes	761	18.2 (16.1–20.4)	21.2 (16.5–26.9)	78.8 (73.1–83.5)	0.03
No	3646	81.8 (79.6–83.9)	15.3 (13.2–17.6)	84.7 (82.4–86.8)	
Mobility					0.48
Yes	1577	35.4 (32.9–38.0)	17.3 (14.4–20.8)	82.7 (79.2–85.7)	
No	2830	64.6 (62.0–67.1)	15.8 (13.4–18.7)	84.2 (81.3–86.6)	
Self-care					0.01
Yes	430	9.8 (8.5–11.3)	24 (18.3–30.9)	76 (69.1–81.7)	
No	3977	90.2 (88.7–91.5)	15.5 (13.5–17.8)	84.5 (82.2–86.5)	
Independent living					0.02
Yes	625	13.7 (12.1–15.5)	22.2 (17.3–27.9)	77.9 (72.1–82.7)	
No	3782	86.3 (84.5–88.0)	15.4 (13.4–17.8)	84.6 (82.2–86.7)	
Disability count					0.17
No	2107	49.6 (46.8–52.4)	14.9 (12.3–18.1)	85.1 (82.0–87.7)	
1 Disability	1077	23.6 (21.3–26)	15.6 (11.7–20.4)	84.4 (79.6–88.3)	
2 Disabilities	572	12.3 (10.6–14.3)	17.9 (13–24.0)	82.1 (76.0–87)	
3+ Disabilities	651	14.5 (12.9–16.4)	21.2 (16.6–26.7)	78.8 (73.3–83.4)	

Abbreviation: LCS, lung cancer screening.

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TABLE 3

Regression models explaining the relationship between disability and lung cancer screening use.

	Model 1		Model 2		Model 3	
Characteristics	**OR (95% CI)**	***p* value**	**OR (95% CI)**	***p* value**	**OR (95% CI)**	***p* value**
Disability						
No	1		1		1	
1 Disability	1.06 (0.71–1.58)	0.782	1.01 (0.67–1.53)	0.953	0.82 (0.54–1.27)	0.376
2 Disabilities	1.35 (0.88–2.07)	0.174	1.35 (0.86–2.11)	0.187	1.03 (0.63–1.71)	0.895
3+ Disabilities	1.69 (1.14–2.49)	0.008	1.69 (1.12–2.54)	0.013	1.17 (0.74–1.85)	0.500
Age						
55–64	1		1		1	
65–74	1.85 (1.37–2.51)	<0.001	1.58 (1.17–2.15)	0.003	1.23 (0.90–1.69)	0.195
75–79	1.12 (0.63–2.02)	0.696	0.92 (0.51–1.65)	0.773	0.72 (0.37–1.41)	0.336
Sex						
Female	1		1		1	
Male	0.95 (0.70–1.29)	0.755	0.97 (0.71–1.33)	0.860	1.14 (0.83–1.57)	0.402
Race and ethnicity						
White	1		1		1	
Black	1.11 (0.48–2.57)	0.804	1.16 (0.50–2.68)	0.731	1.23 (0.55–2.76)	0.614
Other	0.64 (0.34–1.18)	0.150	0.70 (0.37–1.30)	0.254	0.72 (0.38–1.37)	0.316
Marital status						
Never	1		1		1	
Married	1.69 (0.92–3.11)	0.092	1.75 (0.90–3.40)	0.098	1.51 (0.75–3.05)	0.247
Divorced/Separated	1.43 (0.76–2.71)	0.267	1.45 (0.74–2.84)	0.273	1.41 (0.69–2.87)	0.346
Widowed	1.64 (0.81–3.30)	0.168	1.67 (0.80–3.46)	0.172	1.42 (0.66–3.05)	0.364
Education						
Less than high school			1		1	
Attended college			1.20 (0.86–1.67)	0.281	1.15 (0.82–1.61)	0.428
Graduated college			1.11 (0.77–1.61)	0.579	1.09 (0.74–1.61)	0.669
Income						
<$25,000			1		1	

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	Model 1		Model 2		Model 3	
Characteristics	**OR (95% CI)**	***p* value**	**OR (95% CI)**	***p* value**	**OR (95% CI)**	***p* value**
$25,000–<50,000			1.07 (0.74–1.56)	0.709	1.14 (0.79–1.64)	0.495
$50,000–<75,000			0.84 (0.52–1.37)	0.492	1.00 (0.61–1.62)	0.984
$75,000+			0.67 (0.40–1.12)	0.126	0.75 (0.45–1.26)	0.277
Region						
Northeast			1		1	
Midwest			0.89 (0.61–1.30)	0.560	1.03 (0.70–1.49)	0.903
South			0.75 (0.47–1.18)	0.213	0.80 (0.51–1.27)	0.343
West			0.67 (0.40–1.11)	0.117	0.77 (0.47–1.28)	0.315
Metropolitan residency						
No			1		1	
Yes			1.05 (0.76–1.44)	0.775	1.08 (0.78–1.49)	0.648
Insurance						
No			1		1	
Yes			4.83 (1.87–12.46)	0.001	2.94 (1.07–8.04)	0.036
Unmet healthcare needs due to cost in the past 12 months						
No			1		1	
Yes			0.58 (0.35–0.95)	0.031	0.56 (0.34–0.91)	0.019
Routine checkup						
5+ years ago					1	
Past year					7.74 (1.33–44.91)	0.023
Past 2 years					2.68 (0.41–17.39)	0.302
Past 5 years					0.54 (0.04–7.56)	0.647
General health						
Fair-poor					1	
Very good-excellent					0.96 (0.61–1.50)	0.849
Good					0.99 (0.67–1.47)	0.976
Lung disease						
No					1	
Yes					2.53 (1.79–3.56)	<0.001
Asthma						

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	Model 1		Model 2		Model 3	
Characteristics	**OR (95% CI)**	***p* value**	**OR (95% CI)**	***p* value**	**OR (95% CI)**	***p* value**
No					1	
Yes					0.88 (0.57–1.35)	0.551
Cancer history						
No					1	
Yes					2.34 (1.63–3.34)	<0.001
Smoking						
Former					1	
Current					1.00 (0.74–1.34)	0.985

Abbreviation: LCS, lung cancer screening.